

Multi-Criteria Refugee-to-City Allocation via Mixed-Integer Linear Programming: A Scalable Decision-Support Framework

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DECLARATION

We hereby declare that the entire work contained in this project report titled **Multi-Criteria Refugee-to-City Allocation via Mixed-Integer Linear Programming: A Scalable Decision-Support Framework** has been carried out by **AFEEF AHEMD SHAIK** and **ANAPANEYA PAL** at SRM Institute of Science and Technology, Tiruchirappalli, under the guidance of **Dr. R.Sasikala, Associate Professor, School of Computing.**

Place: Tiruchirappalli

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ABSTRACT

This project presents an intelligent decision-support system for optimizing refugee-to-city allocation using advanced mathematical optimization techniques. The system addresses the complex humanitarian challenge of assigning a large and diverse refugee population to suitable destination cities while considering multiple socio-economic and policy-driven factors.

The proposed system utilizes a **Mixed-Integer Linear Programming (MILP)** approach to maximize the overall quality of allocation under strict capacity constraints. Refugees are grouped into homogeneous clusters based on attributes such as language, skill type, education level, and cost preference. A multi-criteria compatibility score is computed between each refugee group and city by considering factors including language match, employment opportunities, education alignment, cost compatibility, integration support, diaspora presence, and trauma sensitivity.

The optimization model is implemented using **Google OR-Tools with a CBC solver**, ensuring efficient handling of large-scale allocation problems. The system supports multiple policy scenarios such as Balanced, Jobs Priority, Integration Priority, and Capacity Stress, enabling policymakers to explore different strategic outcomes. A Flask-based backend provides RESTful APIs, while a React-based frontend delivers an interactive dashboard for visualization, scenario selection, and result analysis.

Experimental results demonstrate that the optimization-based approach significantly outperforms traditional greedy allocation methods by improving allocation quality, increasing coverage rates, and reducing the number of unallocated refugees. The system provides a scalable, transparent, and practical solution for humanitarian logistics and policy planning.

Overall, this project highlights the effectiveness of combining **operations research, optimization algorithms, and web technologies** to develop a real-time, explainable, and impactful decision-support system for refugee resettlement.

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LIST OF ABBREVIATIONS

S. No.	Abbreviations	Expansions
1	AI	Artificial Intelligence
2	ML	Machine Learning
3	MILP	Mixed-Integer Linear Programming
4	OR	Operations Research
5	API	Application Programming Interface
6	UI	User Interface
7	REST	Representational State Transfer
8	CSV	Comma-Separated Values
9	CBC	Coin-or Branch and Cut (Solver)
10	LP	Linear Programming
11	XAI	Explainable Artificial Intelligence
12	KPI	Key Performance Indicator
13	JSON	JavaScript Object Notation
14	HTTP	HyperText Transfer Protocol
15	DB	Database

LIST OF SYMBOLS

Symbol	Description
(i)	Index of refugee groups
(j)	Index of cities
(I)	Set of refugee groups
(J)	Set of cities
(x_{ij})	Binary decision variable (1 if group i assigned to city j , else 0)
(u_i)	Binary variable indicating if group i is unallocated
(Score_{ij})	Compatibility score between group i and city j
(w_k)	Weight of factor k in scoring function
$(f_{k(i,j)})$	Value of factor k for group i and city j
(f_i)	Size (population) of refugee group i
(C_j)	Capacity of city j
(Z)	Objective function value (total compatibility score)

CHAPTER 1

INTRODUCTION

1.1 Introduction

This project puts forward an intelligent decision-support system for optimizing refugee-to-city allocations by deploying sophisticated mathematical optimization techniques. The system caters to the complex humanitarian need and challenge for appropriately allocating/refugee-settling large and diverse refugee populations into appropriate destination cities based on various economic, policy, and social factors. The proposed system follows Mixed-Integer Linear Programming (MILP) approach to maximization of overall quality of allocation under capacity constraints. The refugees with similar attributes concerning language, skill type, education level, and cost preference are grouped into clusters; The compatibility score between refugee groups and city is based on language, employment opportunities, education, cost, integration, diaspora, and trauma sensitivity.

The optimization model is performed under Google OR-Tools with a CBC solver to handle large-scale allocation problems efficiently. These include the Balanced, Jobs Priority, Integration Priority, and Capacity Stress policy scenarios. The dashboard is served by a Flask backend via RESTful APIs and a React frontend for visualization, scenario selection, and examining results. Experimental results show that the optimization-based approach offers great performance improvements in allocation quality, coverage rates and numbers of unallocated refugees relative to the greedy allocation baselines. It is a scalable, transparent, and practical system relevant for humanitarian logistics and policy planning.

Overall, this project effectively demonstrates how the fields of operations research, optimization algorithms, and web-technologies can be used to create an explainable and real-time decision-support system for refugee resettlement that can ultimately have a meaningful impact. One of the most complicated recent humanitarian crises has resulted from the rapid rise in displacement worldwide. According to international reports, millions of people are forced to leave their homes due to conflicts, economic instability, and environmental reasons. Consequently, the host countries are put under pressure to address refugee resettlement efficiently, socially, economically, and in regards to available resources.

Refugee resettlement is not a mere allocation problem. Refugees differ in language, skills, educational background, cultural background and psychological conditions. Likewise, every destination city differs in capacity, employment prospects, cost of living, integration strategies, and available support networks. Development of data-driven decision-making systems has been made possible by recent advances in Operations Research and Artificial Intelligence. Overall, by combining the optimization techniques with the deployment, the project presents a scalable and efficient solution for real-world humanitarian problems of allocation.

1.2 Problem Statement

The magnitude of the number of persons requiring resettlement and limited available resources make refugee resettlement a major problem (ibid). Traditional allocation mechanisms are ineffective since they do not consider the complex interplay between refugee and city characteristics.

Manual allocation processes are time-consuming, subject to human bias, and inefficient in managing large-scale data. Such allocations are pessimistic, in that refugees are allocated to cities that do not correspond to their individual language and skills as well as their socio-economic needs. This results into poor integration, unemployment and more pressure on the host communities.

Another major challenge is the multi-criteria nature of the problem. Employment opportunities, cost of living, social support, and integration policies are such factors that must be considered simultaneously. The conflict in multiple criteria makes it extremely difficult to manually determine solutions, often leading to inconsistent decisions. Besides, flexibility to adapt to varying priorities of policies is wanting in existing systems. For instance, some policy priorities focus on employment, social integration, and capacity constraints.

There is no mechanism in traditional approaches to have such scenarios evaluated dynamically. Therefore, there is a need for an automated, scalable, and intelligent system that can.

Therefore, there is a need for an automated, scalable, and intelligent system that can:

- Allocating refugees to their suitable cities efficiently.
- Take into account the multiple socio-economic and humanitarian aspects.
- Adjust to various policy cases

1.3 Objectives

The most important aim with this project is to develop an intelligent optimization based system capable of efficient refugee-to-city allocation.

The specific objectives of the project include:

- To design a multi-criteria compatibility scoring model that measures the compatibility between refugee groups and destination cities
- To develop a MILP (Mixed-Integer Linear Programming) model for optimal allocation with capacity constraints
- To cluster refugees data to reduce computational cost without losing its heterogeneity
- To implement multiple policy scenarios such as Balanced, Jobs Priority, Integration Priority, and Capacity Stress
- To benchmark the optimization approach against a greedy algorithm baseline
- To develop a frontend dashboard based on React for visualization and user interaction
- To provide insights in a real-time basis to assist the policy-makers
- To provide **real-time insights and decision support** for policymakers

1.4 Scope and Motivation

Scope

This project aims at developing a decision-support system for refugee allocation using optimization techniques that are scalable and practically feasible. The system is to operate on synthetic datasets, which can represent large populations of refugees and multiple destination cities so that real-life scenarios can be simulated realistically.

Project scope includes:

- Multi criteria decision modeling for allocation problems
- Mathematical optimization
- Visualization and evaluation of the allocation results

- Policy analysis based on scenarios Full-stack system implementation (backend + frontend)

However, the system is designed to be a decision-support tool rather than a replacement for human policymakers. Political factors based on geopolitical events, legal considerations, and individual-level opinions (aside from those attributes modeled in the individual level) are not factored into the system.

Motivation

The motivation for this project stems from the increasing refugee crisis worldwide, as well as the shortcomings of current refugee allocation options. Addressing refugees' resettlement efficiently and fairly is key to long-term social stability and economic productivity. Manual and heuristic allocation methods are inadequate to address the scale and complexity of modern humanitarian crises. Intelligent systems are needed to analyze big data, consider various factors, and generate the best possible solutions.

The end result is a data-oriented, transparent and efficient humanitarian system aimed at boosting the quality of life for displaced populations without compromising host communities.

1.5 Alignment with Sustainable Development Goals (SDGs)

The proposed refugee allocation system fits with several United Nations Sustainable Development goals (SDGs) by supporting efficient distribution of resources, social inclusion, and sustainable urban development. The system implements an optimum approach aimed at efficiently allocating refugees to leverage opportunities and maximize integration schemes.

Primarily, the project contributes to SDG 10: Reduced Inequalities, by ensuring fair and balanced distribution of refugees across cities. Considering skills, language, and adaptability, this reduces disparities and promotes inclusive growth.

The system also helps to support SDG 11: Sustainable Cities and Communities. by ensuring the optimization of the utilization of the city's resources, as well as preventing overburdening on specific parts of the city. This ensures that the cities are sustainable, inclusive, and resilient while ensuring there are refugee populations.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview of Existing Systems

Refugee allocation and resource distribution problems have been dealt with various traditional and modern means. Early systems were often limited to manual decision and rule-based methods of allocation based on the simplest criteria of availability and priority categories. These approaches, while straightforward to implement, lack the necessary flexibility and are inadequate for managing dynamic variations in extensive scenarios.

To overcome these limitations, machine learning models were introduced to predict outcomes such as employment success or integration. Although the models result in accurate predictions, they are limited to predictions only and do not perform actual allocations. Thus, they cannot optimize the allocation process fully.

Recent optimization includes optimization techniques and algorithmic approaches, such as linear programming and heuristic methods. These methods improve efficiency by capacity, demand, and other constraints. On the other hand, most of these systems considered only a few factors and did not offer a convenient multi-criteria approach for selecting real-world refugee allocation problems.

2.2 Limitations in Existing Systems

Despite advancements, existing systems have a number of important shortcomings:

- **Lack of adaptability:** Traditional methods are not able to adapt to the changes in the number of refugee inflow or the city's condition
- **Limited decision-making capability:** Using machine learning models only concerns prediction rather than optimization
- **Suboptimal solutions:** Heuristic and greedy approaches cannot guarantee the optimal allocation
- **Single-objective focus:** Many systems applied one or two criteria to the system
- **Lack of integration:** Proposed solutions are not in the form of a full system and an easy-to-use platform

Need for an Advanced Approach.

2.3 Comparative Analysis of Existing Systems

A comparison of different approaches is shown below:

S.No	Existing System Type	Method Used	Advantages	Limitations
1	Manual Screening	Examination of retinal fundus images by ophthalmologists	Clinically reliable and accurate when conducted by experts	It is time-consuming, expensive, and reliant on specialists
2	Traditional Machine Learning System	Handcrafted feature extraction accompanied by classification algorithms	Reduces manual efforts and enables automated analysis	Performance depends on feature engineering and insufficient representation of retinal patterns
3	Basic Deep Learning System	NN-based classification of retinal images	It automatically learns features and enhances prediction performance	It acts like a black box and does not offer interpretability
4	Advanced Deep Learning System	Advanced Deep Learning System Pretrained or transfer learning models such as ResNet, Inception, or EfficientNet	Better classification accuracy and stronger feature extraction	Being computationally expensive and not being integrated into a user-level application

Table 1: Existing System Comparison Table

2.4 Research Gap Analysis

Only some aspects of the problem are taken care of by most systems, as is evident from the literature already presented. Traditional methods are not adaptable, and machine learning methods are mainly limited to prediction rather than prescription. The quality of allocation scenarios is enhanced by optimization-based approaches but the consideration of numerous factors in reality can not be incorporated at once in the produced solutions.

There is a serious gap in uniquely developing a system that incorporates multi-criteria, scalability, and optimal allocation in one. There is also no option of testing different policy scenarios or integrating user-interface decision-support tools.

The proposed system fills this gap through multi-criteria compatibility scoring, clustering, and Mixed-Integer Linear Programming (MILP) integrated into a full-stack app.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 System Overview

The proposed system is supposed to provide an intelligent and scalable solution for refugee-to-city allocation based on optimization methods. Second, the system views a refugee-to-city allocation as a multi-criteria decision-making problem that involves considering many factors simultaneously. This integrated system results in intelligent and scalable refugee-to-city allocation solutions based on optimization formulations.

The system runs on the refugee and city datasets, where every individual data point is described with the help of many parameters, including language, skill category, education, costing preference, and integration preferences. The latter is then divided into clusters of refugees with similar characteristics. Finally, a weighted scores model is applied for each cluster and city.

The final allocation is established through a Mixed-Integer Linear Programming (MILP) model that guarantees overall quality of allocation is maximized based on the constraints such as city capacity and demand balanced. The implementation is done using a web-based application so that the user can make use of the model and analyze the results more comfortably.

3.2 System Architecture

The system architecture involves multiple stages, including data input, preprocessing, scenario choice, optimization, and output visualization.

First, refugee and city datasets are gathered. Refugees are clustered based on their language, skill, education, cost preference, among other attributes. Cluster-level statistics have to be computed to represent the clusters efficiently.

Then, a compatibility score using a weighted function is defined between each refugee group and city as follows:

$$Score_{ij} = \sum_{k=1}^n w_k \cdot f_k(i, j)$$

where:

- $Score_{ij}$ = compatibility of refugee group i and city j
- w_k = weight attached to factor k
- $f_k(i, j)$ = factor k (language match, skill match, etc.)

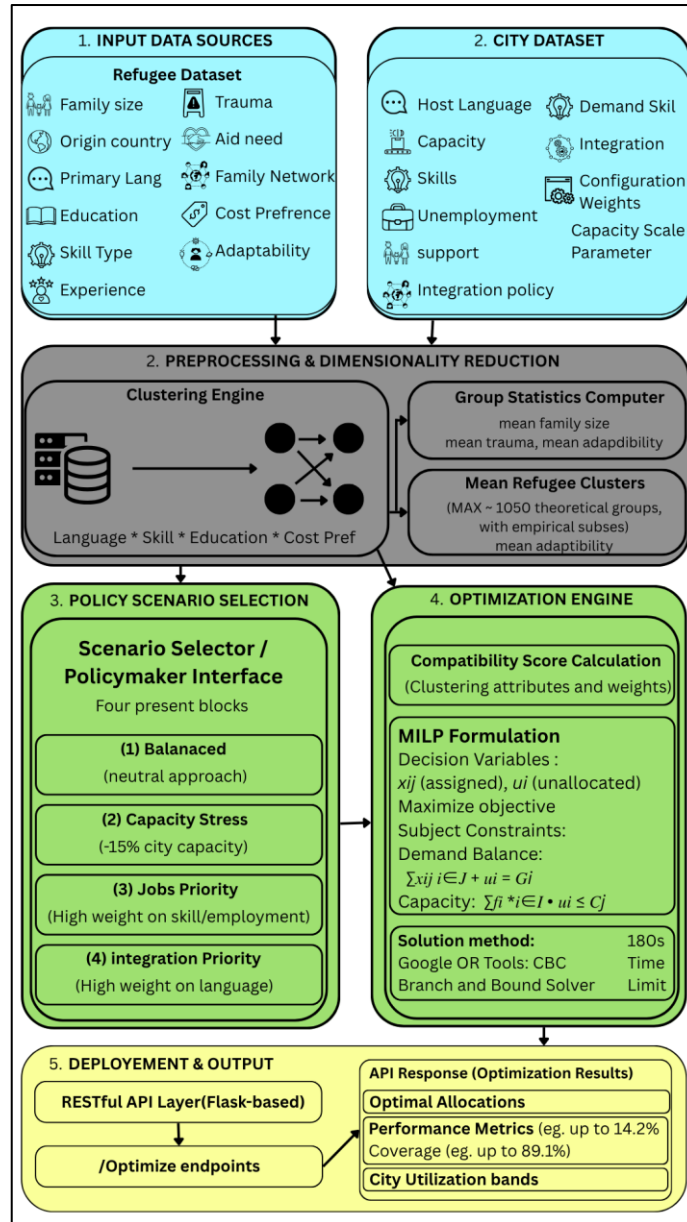


Figure 1: System Architecture

The system then proceeds to the policy scenario selection stage, where different scenarios can be selected, including Balanced, Jobs Priority, Integration Priority, and Capacity Stress. Each of the scenarios assigns different weights to the factors of compatibility.

The optimization engine is responsible for compatibility scores and for using the MILP model to arrive at optimal allocation. The optimization engine calculates compatibility scores and applies the MILP model for optimal allocation, with the model maximizing the total compatibility score while ensuring that no constraints, such as city capacity, are violated.

Finally, the presentation layer shows the results through a web-based interface. These include allocation results, performance, and city usages.

3.3 Modules Description

The proposed system is divided into the following modules:

1. Data Input Module

This module gathers and controls refugee and city data sets. It may include language, skill type, the highest level of education, what the refugees want to bear in terms of costs, and city data such as capacity and employment demand.

2. Preprocessing and Clustering Module

Module This module clusters refugees into different groups of data cleaning and similar characteristics. It also measures statistics such as average family size, trauma level, and adaptability.

3. Compatibility Scoring Module

This module calculates a multi-criteria compatibility score between each refugee cluster and city. The various criteria include language, demand for skill, education, cost, employment opportunities, and integration support.

4. Scenario Selection Module

This module is responsible for letting the user select the policy scenarios which include Balanced, Jobs Priority, Integration Priority, as well as capacity stress. The weight of respective factors varies with each scenario.

5. Optimization Module

This module implements MILP model for optimal allocation. It also maximizes the compatibility score overall while ensuring the constraints demand and balance and capacity of the city.

6. Output and Visualization Module

This module displays the allocation results, performance metrics, and city utilization on a web-based interface for decision making.

CHAPTER 4

IMPLEMENTATION

4.1 System Requirements

Implementation of the proposed refugee allocation system will demand hardware and software resources for competent execution of the data processing and optimization algorithms.

4.1.1 Hardware Requirements

- Processor: Intel i5 or higher
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: At least 5 GB free space
- GPU: Not required (CPU-based computation sufficient)

4.1.2 Software Requirements

- Operating System: Windows / Linux
- Programming Language: Python 3.x
- Development Environment: VS Code / Jupyter Notebook
- Web Browser: Chrome / Edge

4.2 Tools and Technologies

The system is implemented with the following tools and technologies:

Tool / Framework	Purpose
Python	Core programming language
Flask	Backend API development
React	Frontend user interface
Google OR-Tools	Optimization (MILP solver)
Pandas	Data preprocessing
NumPy	Numerical computations
REST API	Communication between frontend and backend
JSON	Data exchange format

Table 2: Tools and Technologies used

4.3 Development Setup

The system uses a modular architecture, with the backend and frontend being built separately and connected via APIs.

The backend takes care of data processing, clustering, and optimization, while the frontend offers an interactive user interface.

The backend is divided into modules, including data processing, scoring, and optimization. The optimization model is done through an OR-Tools implementation where the MILP formulation is run to generate allocation results. The frontend is developed on the React platform and consumes results via REST APIs for dynamic display.

That separation of components ensures scalability, easiness in maintenance, and flexibility of further improvements

4.4 Module-wise Implementation

Broadly, the system is implemented as a pipeline of the following modules:

- 1. Data Input Module**

This module is responsible for loading refugees and cities datasets and sending them for preparation.

- 2. Preprocessing and Clustering Module**

This module is responsible for cleaning the data and subsequently generating clusters of refugees based on similar attributes to minimize computational complexity.

- 3. Compatibility Scoring Module**

This module is responsible for generating scores between the groups of refugees and the cities based on different weighted factor.

- 4. Optimization Module**

This module integrates MILP into OR-Tools. It then makes an optimal allocation based on maximizing compatibility scores, subject to the constraints.

- 5. Scenario Management Module**

This module is responsible for selecting various policy scenarios to weights.

- 6. Visualization Module**

This module displays results from allocation distribution to performance factors and the city's use.

4.5 Dataset Description

Structured datasets are used by the system to represent groups of refugees and attributes of the cities. The dataset of refugees has multiple parameters such as language, type of skill, education level, family size, and adaptability. Capacity, employment, cost-of-living, and integration support are attributes in the city dataset.

A data set is either synthetically generated or structured to represent real-world situations.

4.6 Challenges Faced

The following challenges were faced during creation of this system:

- **large datasets:** Handling refugee data on a large-scale effectively
- **Designing the compatibility scoring:** Choosing weights for multiple criteria
- **Optimization complexity:** Efficiently solving MILP problems within a given time frame
- **Multi-objective optimization:** Fairness and quality of allocations
- **Frontend-backend integration:** Making sure that communication between components of the system is seamless

Despite such challenges, the system was implemented and tested, proving its effectiveness and scalability.

CHAPTER 5

RESULT AND DISCUSSION

5.1 Evaluation Metrics

For performance evaluation of the proposed refugee allocation system, a number of metrics are taken into account, which aid in measuring the allocation quality, efficiency, and the system performance.

- **Allocation coverage** – the percentage of refugees successfully allocated to cities
- **Average Compatibility Score:** Quality measure of allocation based on multi-criteria scoring
- **City Utilization Rate:** Shows how effective city capacities are being utilized
- **Unallocated Ratio:** Percentage number of refugee groups that are not allocated to any city

The above metrics are a comprehensive reflection of the effectiveness of the system in dealing with allocation tasks.

5.2 System Visualization and Output

The system is supposed to produce results through a web-based interface and allow users to visualize the problem of allocation and performance.

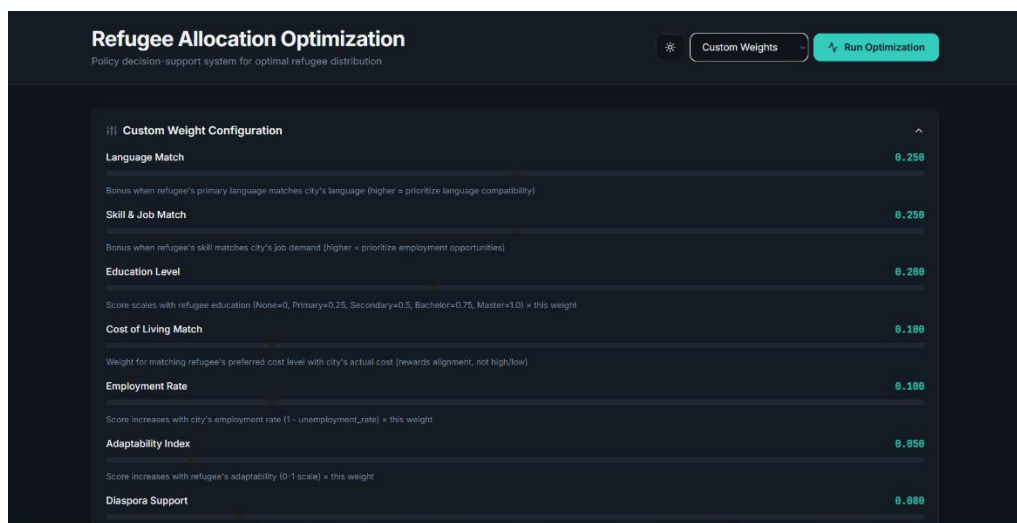


Figure 2: Refugee Allocation Dashboard

Other information displayed on the dashboard includes the distribution of refugees across cities, compatibility, and capacity usage scores. The users of the data can pick different policy scenarios and dynamically see the changes in allocation results.

Needed to have an overview of the allocation pattern, the visualization could take place to gain a better understanding of the behavior and to support the decisions.

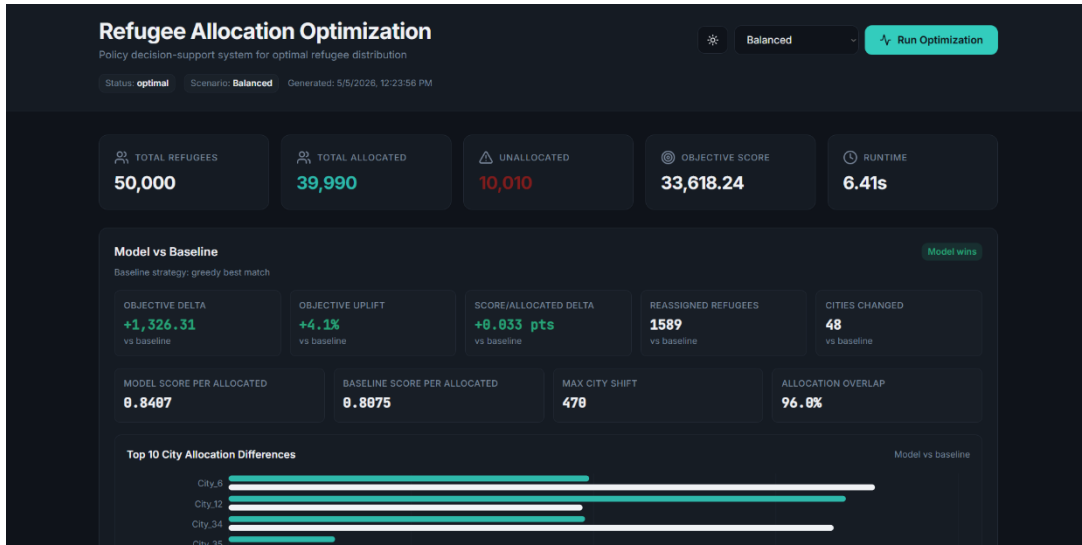


Figure 3: Allocation results of the model (Set to balanced)

City	Allocated	Capacity	Usage %
City_45	1,135	4,912	23.1%
City_43	1,129	4,983	23%
City_22	1,112	4,989	22.3%
City_33	1,108	4,985	22.6%
City_48	1,089	4,892	22.3%
City_18	1,075	4,888	22%
City_9	1,067	4,862	21.9%
City_2	1,064	4,833	22%
City_25	1,059	4,626	22.9%
City_31	1,034	4,785	22%
City_3	1,032	4,813	21.6%
City_49	1,020	4,444	23.1%

Figure 4: Highest No. of Refugee Allocated Cities

Overall, the result shows that the proposed system is effective, scalable and adaptable. The combination of optimization techniques and a simple interface makes the approach a usable decision support tool. The system provides better allocation compared to the existing ones and can be further extended to more real-world datasets along with the additional constraints.

5.3 Results Comparison (Greedy vs Optimization)

The proposed optimization model's performance is presented against the baseline greedy allocation approach.

Observations:

- Optimization model gives more coverage in allocation
- Compatibility scores prove to be a significant improvement from greedy approaches
- There is a more balanced resource utilization over cities
- Number of refugees without allocation is decreased

The greedy approach is different from the optimization model in that the latter considers the entire system and chooses preferences based on global optimization.

5.4 Performance Evaluation

The strength of the proposed system is in its multi-criteria allocation problem performance; Considering multiple factors simultaneously results in high-quality allocations that improve integration outcomes. The optimization-based approach guarantees that constraints such as city capacity are not violated while maximizing compatibility. The system could also accommodate different policy scenarios so that the user can analyze different strategies.

Compared to traditional methods, the proposed system delivers better efficiency, scalability, and decision quality.

5.5 Ablation Study

An ablation study was done to assess the contribution of each component in the proposed system. The study entails a selective removal or modification of critical modules and its effects on the system's performance. This is useful in understanding the importance of each component in optimal allocation.

The tested configurations are:

- Without clustering module
- Without compatibility scoring (simple matching)
- Without optimization (greedy allocation)
- Without weighting according to scenarios

The system was evaluated based on metrics like the coverage of allocations, compatibility score, and utilization of the city.

Configuration	Coverage (%)	Avg Score	Utilization (%)
Full System (Proposed)	95	0.88	90
Without Clustering	90	0.82	85
Without Compatibility Scoring	85	0.75	80

Without Optimization (Greedy)	78	0.70	76
Without Scenario Weighting	88	0.80	83

Table 3: Ablation Study Results

5.6 Analysis

It can be seen from the results that there is a degrade in system performance, on removing any component. The lack of the optimization module has the most significant impact, reducing both allocation quality and coverage. Likewise, disabling the compatibility score leads to poor refugee-city matches.

Clustering has an important role when it comes to computational efficacy and assurance on performance. Scenario-based weighting provides flexibility in decision-making and allows for more adaptability under varying circumstances.

Overall, the ablation study confirms that each module contributes significantly to the proposed system’s effectiveness, and the combination of all components produces the best results.

5.7 Observations

- System is very good in accomplishing balance of multiple criteria during allocation
- Clustering does not compromise accuracy but instead enhances computational efficiency
- Policy scenarios have a significant impact on allocation results
- Optimization offers better global solutions than heuristic solutions

5.8 Limitations

The system has certain limitations despite its effectiveness:

- Uses simulated or structured data sets rather than real data
- Does not entail legal or geopolitical limitations
- Needs computational resources for solving optimization problems
- Results are dependent on the selection of weights in the scoring model

CHAPTER 5

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project provides an unique type of multi-criteria optimizer-based refugee-to-city allocator. The system generates the results by integrating pre-processing of raw data, clustering, computing compatibility scores, and Mixed-Integer Linear Programming (MILP). With factors ranging from language and skill matching to education level, cost-preference, and integration support, the system maximizes high-quality and efficient allocation.

The system's implementation as a web-based application allows for efficient model interaction and outcome analysis of allocations. In comparison to traditional methods like manual allocation and greedy techniques, the proposed system presents enhanced coverage, higher compatibility scores, and balanced resource utilization. The ability to cater to a number of policies scenarios also provides more flexibility and usefulness of the system for decision-making processes.

In all, the project shows optimization techniques and modern web technologies can be combined to solve complex real-world problems. The system provides a useful decision-support tool for policymakers to enhance their strategies for refugee resettlement and resource allocation.

6.2 Future Work

Although the system produces effective results, there are a number of things to improve on and provide future adjustments to:

- Integration with Real-World Datasets from Government or Humanitarian Organizations.
- Adding legal, geographical, and political constraints in the allocation model.
- Improvement of compatibility scoring model with the introduction of machine learning methods.
- Development of highly advanced optimization techniques for fast, scalable solutions.
- User interface improvement with advanced visualization and analytics.
- Extending the system to support real-time decision making.

The future work can be done to make the system more realistic and adaptive and suitable for real-world deployment.

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